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Palaeomagnetic study of some charnockites of the Palni Hills

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The intensity of natural remanence magnetisation (NRM) is measured along the three mutually-perpendicular directions using an astatic magnetometer. The intensity of induced magnetisation is measured using an apparatus fabricated in the laboratory. The Koenigsberger ratio has been calculated for all the samples and the ratio has been used to test the stability of NRM in the samples. The mean direction of magnetisation is determined from the three components of the NRM intensity. The palaeomagnetic pole position of the samples is determined using the direction of magnetisation and the site location. An attempt has been made to fix the geological age of the charnockites using palaeomagnetic methods.

Introduction

Generally, rocks contain very minute quantities of magnetic minerals, and by virtue of these materials they acquire magnetic properties. The magnetism acquired by these rocks at the time of their formation remains virtually intact over time and is therefore of great importance in the palaeomagnetic study of rocks, throwing light on the history of geomagnetic field, the possible mechanisms of continental drift and the age of the rock formations. Such studies on rocks of Indian origin have been done extensively by Athavale et al. (1970) and Bhimasankaran and Pal (1970). Their studies have been confined to rocks of Upper Cretaceous to Miocene age. In the present paper, an attempt has been made to fix the geological age of the charnockites of the Palni Hills by measuring their palaeomagnetic pole position and also the stability of magnetisation by determining the Koenigsberger ratio.

1.1. Natural remanence magnetisation

The magnetisation acquired by rocks at the time of their formation is called the natural remanence magnetisation (NRM). The initial magnetisation during the rock formation is due to Earth's magnetic field, and nothing else can possibly be responsible for the uniform magnetisation of a mass of such a large scale. The direction of NRM is fairly uniform, but its intensity varies from point to point even within a rock mass of few cubic metres. A clue to the age of the rock formation can be obtained from the direction and the palaeomagnetic pole position of the remanent magnetisation.

1.2. Koenigsberger ratio

Koenigsberger (1938) introduced a ratio, Q_n, which may be used to study the palaeomagnetic characteristics of rocks. This is defined as the ratio

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of intensity of natural remanence magnetisation, J_n, to that of the induced magnetisation in a field of 0.5 Oe. Likhite and Radhakrishnamurthy (1966) have suggested that the Koenigsberger ratio can be used to test the stability of magnetic directions of rocks. They have suggested, in general, that Q_n values varying between 3 and 10 for igneous rocks are magnetically stable. This technique has been successfully utilised by Bhalla et al. (1974), to test the stability of magnetisation of Deccan trap basalts.

2. Experimental

The intensity of NRM, J_n, was measured in 16 samples of charnockites of Palni Hills using an astatic magnetometer of sensitivity 4.2×10^{-5} Oe cm^{-1} deflection at one metre optical path. The samples were placed along the three possible orientations and the components of the intensity along x, y and z directions were measured. The resultant intensity, J_n, was calculated for all the samples and results are summarized in Table I.

The induced intensity of magnetisation, J_i, of all these samples in a field of 0.5 Oe was measured using an apparatus fabricated in the laboratory. The values of induced magnetisation and Q_n values are given in Table I.

All the samples studied were subjected to a.f. demagnetization to remove secondary magnetizations. This procedure and data on Koenigsberger ratios provide the evidence for stability of the NRM values used in the interpretation.

TABLE I

The NRM intensity (J_n), induced intensity of magnetisation (J_i) and Koenigsberger ratio, Q_n of charnockites from the Palni Hills

Specimen No.	J_n (emu g^{-1})	J_i (emu g^{-1})	$Q_n = J_n/J_i$
113	1.50×10^{-2}	2.57×10^{-3}	5.85
33	2.92×10^{-2}	6.86×10^{-3}	4.23
99	5.19×10^{-3}	1.48×10^{-3}	3.51
195	3.49×10^{-2}	4.58×10^{-3}	7.64
111	7.24×10^{-3}	7.39×10^{-4}	9.80
93	7.97×10^{-3}	2.29×10^{-3}	3.48
91	1.49×10^{-2}	5.46×10^{-3}	2.72
12	5.38×10^{-3}	5.99×10^{-4}	8.98
4	3.79×10^{-3}	7.39×10^{-4}	5.14
148	5.79×10^{-3}	5.28×10^{-4}	10.98
11c	7.82×10^{-3}	8.09×10^{-4}	9.67
73	5.11×10^{-2}	1.47×10^{-2}	3.47
10	1.45×10^{-2}	2.50×10^{-3}	5.80
25	3.68×10^{-3}	1.41×10^{-3}	2.61
123	5.19×10^{-3}	8.80×10^{-4}	5.92
95	7.69×10^{-4}	5.36×10^{-5}	14.35

3. Results

From Table I, it can be seen that the Q_n ratio, varies between 3 and 10, except for one sample whose value is 14. Radhakrishnamurty and Mishra (1966) have reported Q_n values varying between 2 and 6 for quartzites. Verma et al. (1974), have also reported Q_n values for Deccan trap basalts varying from 5 to 10. The value for Q_n obtained for charnockites of igneous origin is also in the range 3 and 10, which is also in accordance with the values of Q_n already obtained for other igneous rocks as mentioned above. Hence, following Radhakrishnamurty and Mishra (1966) who have suggested use of Q_n values for assessing stability condition, it can be concluded that the charnockites under study are palaeomagnetically stable and fit for palaeomagnetic investigations. It should be pointed out, in this context, that no abnormal value for Q_n is obtained in this study.

4. Age of the rocks

The geological age of the rocks can be attempted by finding out the palaeomagnetic pole positions of the rocks. The components of NRM along the three possible orientations were determined using the astatic magnetometer. The inclination, I, and the declination, D, were calculated for the individual samples collected at various altitudes using the procedure given by Tarling (1971) and they are summarized in Table II. The resultant direction was computed and the value is $-35.6°$ inclination and 236.4° declination. The site location of the Palni Hills is 10°14'N and 77°28'E. Following the procedure given by Tarling

Table II

Inclination (I) the declination (D) the palaeomagnetic pole position and altitudes (m) of charnockite samples of the Palni Hills

Sample No.	I (deg.)	D (deg.)	Latitude (°N)	Longitude (°E)	Altitude (m)
123	−39.6	239.1	38.1	50.9	870
195	−45.4	235.1	35.7	51.1	850
73	−41.1	236.4	39.1	51.5	1050
95	−39.9	229.6	45.9	56.9	1250
99	−31.2	234.2	46.9	56.5	1280
91	−33.9	237.5	43.1	54.1	1380
93	−34.4	240.8	40.7	52.6	1380
33	−26.2	233.7	50.9	58.9	2165
10	−36.3	241.1	39.2	51.6	—
4	−27.3	236.3	48.4	57.4	—

(1971), the palaeomagnetic pole position of the individual sample collected at various altitudes was calculated and they are reported in Table II. The mean palaeomagnetic pole position was also calculated using the resultant direction of magnetisation and it is given in Table III along with the palaeopole position of rocks of known age. It was found that the mean pole position is 42.8°N and 54.2°E.

Tarling (1971) has suggested that rocks can be dated using palaeomagnetic methods of determining the mean palaeomagnetic pole positions showing variations of 10^6–10^9 y. According to him, the geological dating can be done by comparing the mean palaeomagnetic pole position of the rock under study with that of the rocks of known age. Tarling (1971) fixed the age of rocks of Czechoslovakia by comparing them with known Czechoslovakian pole positions, and gave the age of the order less than 20–175 Ma. This method was used to attempt the establishing of a geological age for charnockites of the Palni Hills.

Vinagradov et al. (1964), using the lead isochron method and Pb^{207}/Pb^{208} isotope ratio, assigned the age of 2.6 Ga to charnockites of Visakhapatnam. Bhimasankaram (1964) has reported the pole position of Visakhapatnam charnockites as 35°. Subba Rao and Radhakrishnamoorthy (1979) used the pole position of Visakhapatnam charnockites reported by Bhimasankaram (1964) to date the quartz-magnetite from Kistatraopalli and Madampur. They assigned the age of about 2.6 Ga for quartz-magnetite.

The mean pole position of charnockites of Palni Hills and the pole positions of rocks of known age are given in Table III. It is seen from Table III

TABLE III

Palaeomagnetic pole position of charnockites and other rocks belonging to the same age

Rock type	Location	Mean palaeopole position		Reference
		Latitude	Longitude	
Charnockites	Visakhapatnam	35°00′N	–	1964
Quartz-magnetite	Kistraopalli	47°27′N	29°48′E	1979
Quartz-magnetite	Makadampur	47°24′N	26°11′E	1979
Charnockites [a]	Palni Hills	42°48′N	54°12′E	–

[a] Samples investigated in the present study.

that the mean pole position calculated for charnockites of Palni Hills is comparable to the pole position of the rocks of known age. Hence, it may be concluded that the charnockites of Palni Hills may be assigned a geological age of ~ 2.6 Ga.

The palaeomagnetic pole position of individual samples of charnockites collected at various altitudes of Palni Hills is reported in Table II along with the altitudes at which the samples were collected. An inspection of Table II suggests that the samples may be broadly arranged into two groups, depending on the value of palaeopole position of the samples and their altitudes. One group possesses the palaeopole position around 36°N, having an altitude of less than 1100 m, and the other possesses the palaeopole position around 46°N having an altitude greater than 1100 m. It has been pointed out by Tarling (1971) that a change in pole position of $\sim 0.3°$ corresponds to 1 Ma. The change in palaeopole position may be used as a significant criterion in determining the age of formation of a rock. It is quite likely that the two pole positions may represent two different stages of rock formation in the Palni hills. As the mean pole position of 42.8°N corresponds to 2.6 Ga, it is very likely that these rocks would have been formed within a span of ± 20 Ma. As such, the rocks would not have been formed earlier than 2.7 Ga. Such an attempt to classify the rock of Palni Hills under different age group, has already been made by Ramamoorthy (1980). His observation is made on the percentage of cobalt content of the samples collected at various altitudes of Palni Hills. He classified the samples at altitudes of less than 2000 m as being older than the samples having higher altitudes. The present results confirm Ramamoorthy's findings, and are also consistent with Wadia's (1975) observation that the Palni Hills belong to the Archean, i.e. rocks more than 2 Ga old belonging to the Precambrian of India.

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References

Athavale, R.N., Verma, R.K., Bhalla, M.S. and Pullaiah, G., 1970. Drift of the Indian sub-continent since Precambrian times. In: Palaeophysics, Proc. NATO Adv. Stud. Inst., Academic Press, p. 291.

Bhalla, M.S., Mital, G.S. and Murti, C.S., 1974. Application of low-field hysteresis and memory-phenomenon studies in determining the stability of magnetisation of Deccan trap basalts from Mount Girnar and Mount Paragash regions, India. Phys. Earth Planet. Inter., 8: 352–360.

Bhimasankaram, V.L.S. and Pal, P.C., 1970. Palaeomagnetic studies in India–a review. Proc. 2nd Symp. Upper Mantle Project, p. 223.

Bhimasankaram, V.L.S., 1964. A preliminary investigation on the palaeomagnetic direction of the charnockites of Andhra pradesh. Curr. Sci., 33: 465.

Koenigsberger, J.G., 1938. Terr. Mag., 3: 119.

Likhite, S.D. and Radhakrishnamurty, C., 1966. Initial susceptibility and constricted Rayleigh loops of some basalts. Curr. Sci., 35: 534–536.

Radhakrishnamurty, C. and Mishra, D.C., 1966. Bull. N.G.R.I., 4, 103.

Radhakrishnamurty, C., Likhite, S.D. and Sahasrabudhe, 1978. In-situ magnetic measurements on igneous rock bodies. Proc. Indian Acad. Sci., Ser. A, 87: 245–254.

Ramamoorthy, N., 1980. Ph.D. Thesis, Annamalai Univ., Annamalainagar.

Subba Rao, Y.V. and Radhakrishnamurty, I.V., 1979. Palaeomagnetism of quartz-magnetite rocks, Karim Nagar, India. J.R. Astron. Soc., 58: 517–522.

Tarling, D.H., 1971. Principles and Applications of Geomagnetism. Chapman and Hall, p. 73.

Verma, R.K., Mital, G.S. and Bhalla, M.S., 1974. Magnetic properties of Deccan traps from Mount Girnar and Mount Paragash, Gujarat, India. Geophys. Res. bull., 12: 37.

Vinogradov, A., Tugarinov, A., Zhykov, C., Stapnikova, N., Bilikova, E., and Khorre, K., 1964. Geochronology of Indian Precambrian. Proc. 22nd Int. Geol. Congr. Vol. 10, p. 533.

Wadia, D.N., 1975. Geology of India. McGraw-Hill, India.